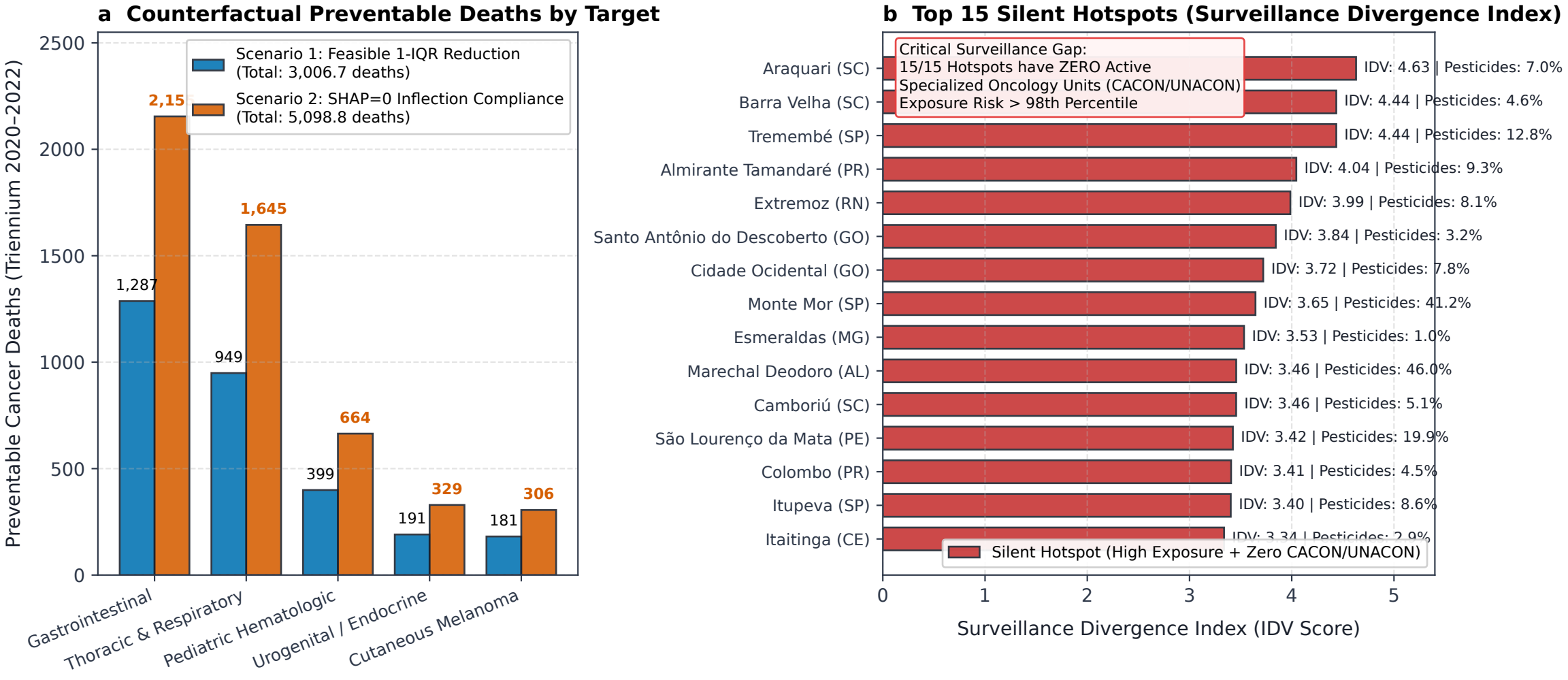
a Counterfactual Preventable Deaths by Target
Preventable Cancer Deaths (Triennium 2020–2022)
Scenario 1: Feasible 1-IQR Reduction
(Total: 3,006.7 deaths)
Scenario 2: SHAP=0 Inflection Compliance
(Total: 5,098.8 deaths)
2,15
1,645
664
329
306
1,287
949
399
191
181
Gastrointestinal
Thoracic & Respiratory
Pediatric Hematologic
Urogenital / Endocrine
Cutaneous Melanoma
b Top 15 Silent Hotspots (Surveillance Divergence Index)
Critical Surveillance Gap:
15/15 Hotspots have ZERO Active
Specialized Oncology Units (CACON/UNACON)
Exposure Risk > 98th Percentile
Araquari (SC)
Barra Velha (SC)
Tremembé (SP)
Almirante Tamandaré (PR)
Extremoz (RN)
Santo Antônio do Descoberto (GO)
Cidade Ocidental (GO)
Monte Mor (SP)
Esmeraldas (MG)
Marechal Deodoro (AL)
Camboriú (SC)
São Lourenço da Mata (PE)
Colombo (PR)
Itupeva (SP)
Itaitinga (CE)
IDV: 4.63 | Pesticides: 7.0%
IDV: 4.44 | Pesticides: 4.6%
IDV: 4.44 | Pesticides: 12.8%
IDV: 4.04 | Pesticides: 9.3%
IDV: 3.99 | Pesticides: 8.1%
IDV: 3.84 | Pesticides: 3.2%
IDV: 3.72 | Pesticides: 7.8%
IDV: 3.65 | Pesticides: 41.2%
IDV: 3.53 | Pesticides: 1.0%
IDV: 3.46 | Pesticides: 46.0%
IDV: 3.46 | Pesticides: 5.1%
IDV: 3.42 | Pesticides: 19.9%
IDV: 3.41 | Pesticides: 4.5%
IDV: 3.40 | Pesticides: 8.6%
Silent Hotspot (High Exposure + Zero CACON/UNACON)
Surveillance Divergence Index (IDV Score)